

14. Answer: D

Considering both waves travelling to the right.

In the first possible scenario, the dotted curve leads the solid curve by $\frac{1}{4}T$, i.e. a phase

difference of $\frac{1}{4}T \times 2\pi = \frac{\pi}{2}$ rad. This option is not given.

In the second possible scenario, the dotted curve leads the solid curve by $\frac{3}{4}T$, i.e. a phase

difference of $\frac{3}{4}T \times 2\pi = \frac{3\pi}{2}$ rad. Hence, the answer.